

Computer Graphics (CSC214)

Unit - 01: Introduction of Computer



Graphics

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Credit Hours: 3hrs

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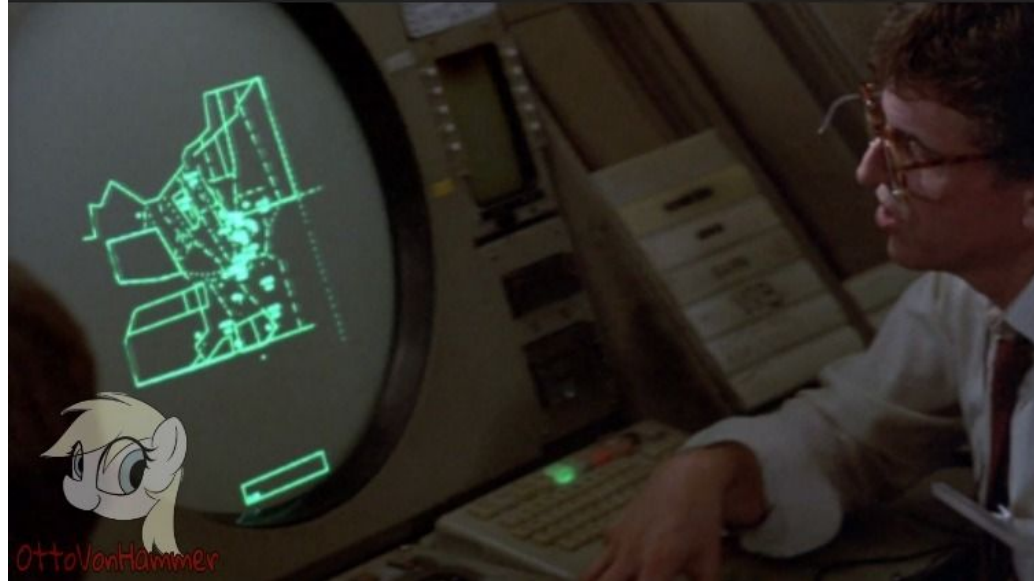
► What is Computer Graphics ?

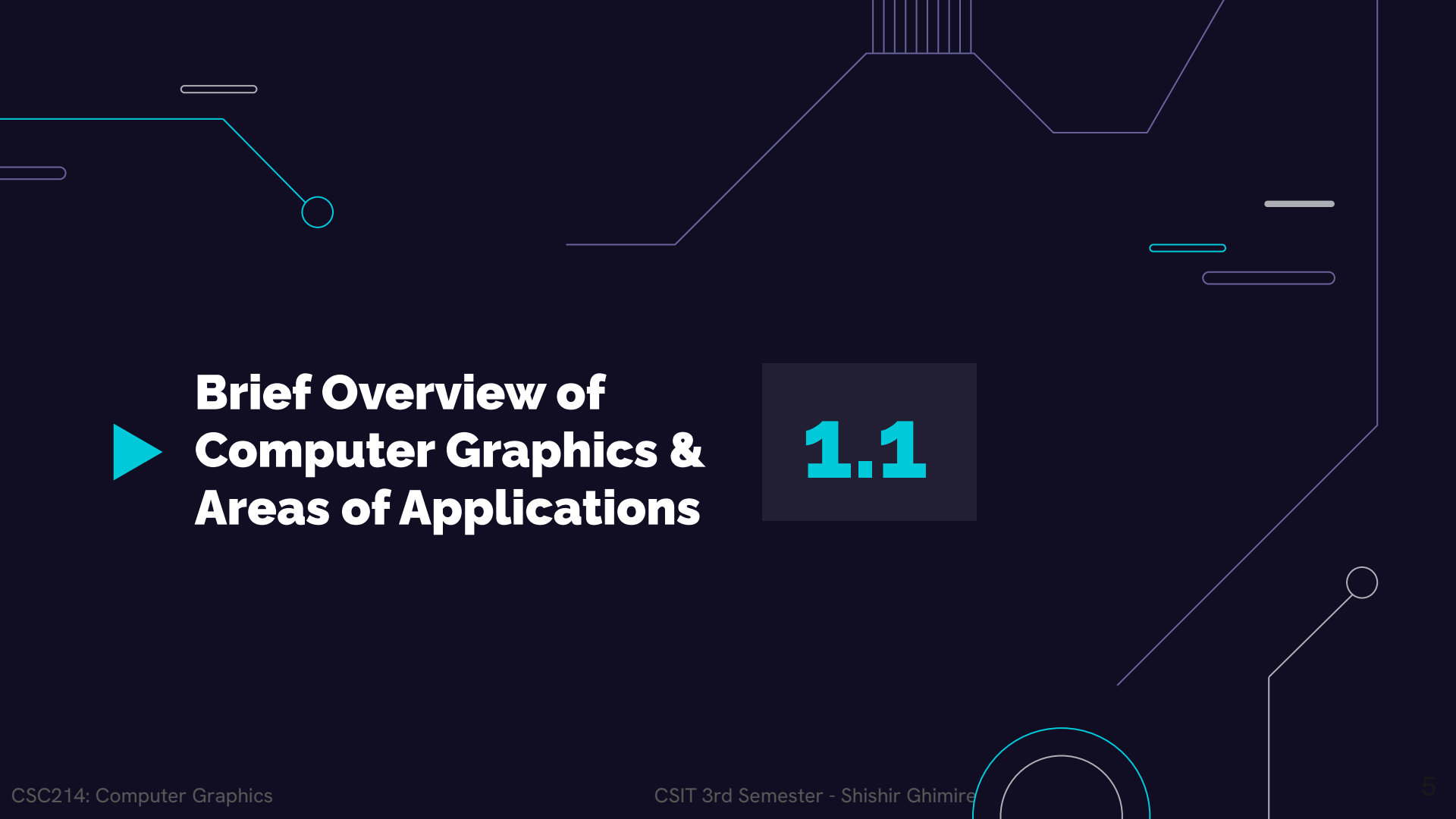


► **Why**
Computer
Graphics ?

"Man never went to the moon, it
was all computer graphics"

Computer graphics in 1969:





Brief Overview of Computer Graphics & Areas of Applications

1.1

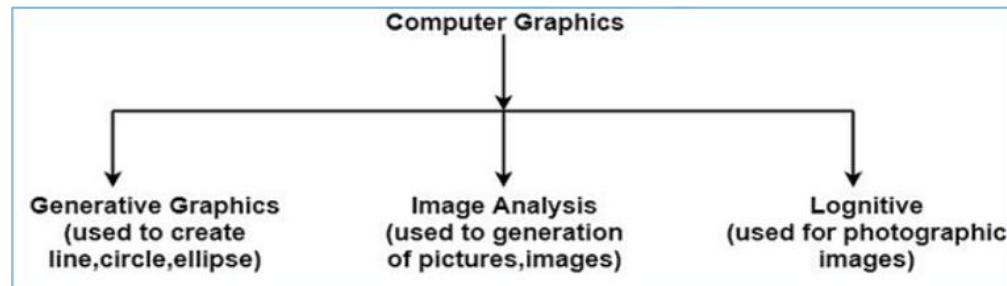
► INTRODUCTION

Computer Graphics is a field related to the **generation of graphics** using computers. It includes the creation, storage, and manipulation of images of objects.

- Computer graphics today is **largely interactive**, i.e. the user controls the **contents, structure, and appearance** of images of the objects by using input devices, such as keyboard, mouse, or touch-sensitive panel on the screen.

Computer graphics refer different things in different contexts:

- **Pictures**, scenes that are generated by a computer.
- **Tools** used to make such pictures, software and hardware, input/output devices.
- The whole field of study that involves these tools and the pictures they produce.



► Main Tasks in Computer Graphics:

❖ Imaging:

- **Formation of an Image:** This involves creating or capturing 2D images.
- **Representation of 2D Images:** Techniques and processes used to represent and manipulate images in two dimensions.

❖ Modeling:

- **Representing 3D Images:** Creating three-dimensional representations of objects or scenes.

❖ Rendering:

- **Constructing 2D Images from 3D Models:** The process of generating 2D images from 3D models by simulating light transport to achieve realistic effects.

❖ Animation:

- **Simulating Changes Over Time:** This involves creating the illusion of movement by displaying a sequence of images or frames.
- **Describing How Objects Change in Time:** Techniques to represent how objects move, deform, or change their appearance over time.

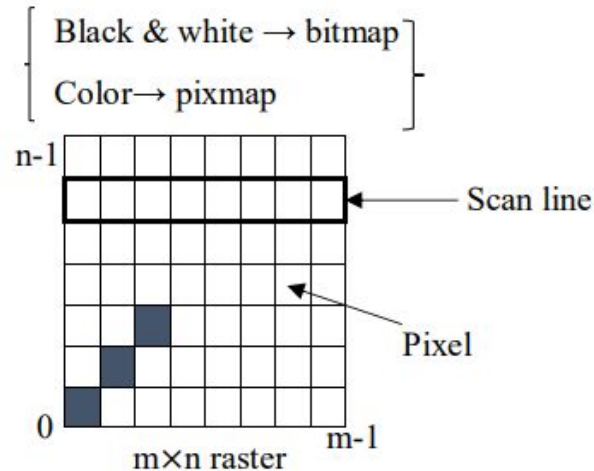
► Basic Concepts:

Raster: A rectangular array of points and dots.

Pixel (picture element): One dot or picture element of the raster.

Scan line: A row of pixel.

Bit map: Ones and Zeros representation of rectangular array of point on the screen.

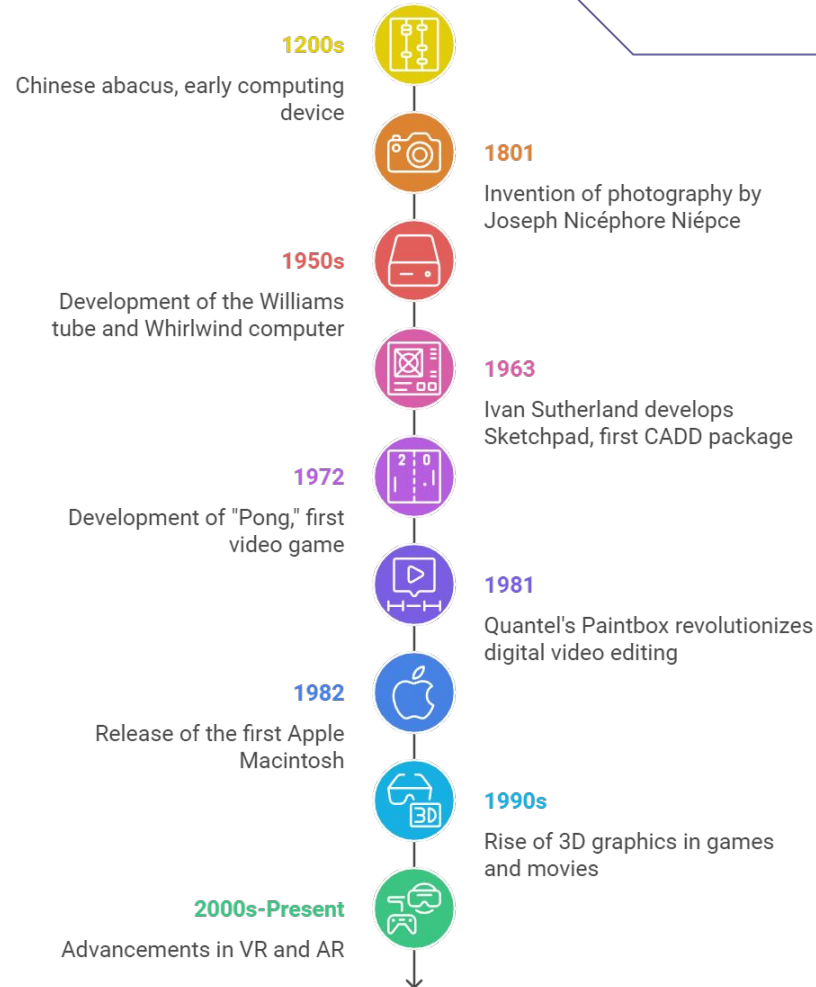


- 8 bit color mode can represent $2^8=256$ different colors.

Note: we don't have any pixel like 1.2, 5.8.

Raster device co-ordinate can have only integer values.

► Historical Evolution of Computer Graphics:



► Historical Evolution of Computer Graphics:

Whirlwind: early graphics using Vector Scope (1951)



Space wars: first computer graphics game (MIT 1961)



first CAD system (IBM 1959)



SketchPad: first interactive graphics (1961)

► Historical Evolution of Computer Graphics:

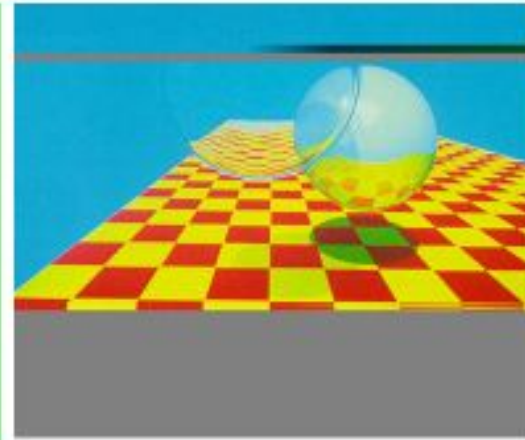


First bump-mapped images (Blinn 1978)

Early texture-mapped image (Catmull 1974)



First distributed ray traced image (Cook 1984)



First ray traced image (Whitted 1980)

► Historical Background (Details):

Year	Milestone
1940	First directly digital computer-generated graphics, first radiosity image invented at MIT.
1950	John Whitney Sr. invents computer-assisted mechanisms for graphic artwork and short films.
1951	Vectorscope computer graphics display on the computer at MIT.
1955	Sage system uses the first light pen as an input device at MIT Lab by Bert Sutherland.
1957	National Bureau of Standards uses the first image-processed photo.
1960	William Fetter first termed "Computer Graphics" for cockpit drawing.
1962	Ivan Sutherland at MIT creates a man-machine graphical communication system.
1965	IBM 2250, the first graphics computer, becomes available.
1966	Ralph Baer develops the first consumer computer game, "Odyssey."
1968	Tony Pritchett creates the first computer animation "FLEXIPEDE" in the UK.
1972	Nolan Bushnell develops Pong, the first electronic game.
1973	Z-buffer algorithm and texture mapping developed by Edwin Catmull.
1974	Phong shading method developed by Phong Bui-Tuong.
1976	First 3D appearance film created by Gary Demos and John Whitey Jr. at Triple-I.
1979	Ray tracing created at Bell Laboratory & Cornell University.

► Historical Background (Details):

Year	Milestone
1981	Computer graphics for IMAX by Nelson Max at Lawrence Livermore National Laboratory; Donkey Kong video game introduced by Nintendo.
1982	First broad use of 3D graphics animation in Disney feature film; AutoCAD 1.0 launched.
1985	Medical imaging software combined with Voxel technology.
1987	Video graphics array (VGA) standard introduced.
1989	Super video graphics array (SVGA) recommended; Tim Berners Lee develops the first website with the original URL.
1993	Mosaic web browser released by UIUC; first public cell phone call made.
1994	Netscape founded by developers of Mosaic.
1995	First fully CGI (Computer-generated imagery) movie released; MS Internet Explorer 1.0 released.
2000	First web-based CAD system, Sketchup, released.
2006	Google acquires Sketchup.
2013	Graphics creation becomes possible on home computers.
2015	Big data used to create animations.
2018	Realistic graphics creation on mobile phones; real-time CGI-based human face creation.

► FUTURE PREDICTIONS:

- ❖ **Widespread Adoption of High-Definition TV (HDTV):** Enhanced display technology will bring sharper images and richer colors to television, offering immersive viewing experiences.
- ❖ **Ubiquitous Wireless Computing:** Increasingly interconnected devices will enable seamless access to information and services from anywhere, anytime.
- ❖ **Advancements in Computer Vision:** Breakthroughs in visual perception technology will empower machines to interpret and understand visual information with human-like accuracy, revolutionizing industries like autonomous vehicles and robotics.
- ❖ **Improvements in Realistic Rendering:** Advancements in graphics technology will lead to more lifelike virtual environments, with realistic rendering of complex materials and enhanced lighting effects, enriching experiences in gaming, animation, and virtual reality.
- ❖ **Potential Shifts in Programming Language Preferences:** Emerging languages and frameworks may reshape graphics programming practices, driven by factors like ease of use, performance, and compatibility with evolving hardware and software platforms.

► APPLICATIONS OF CG :

❖ User Interfaces:

- Used in desktop windows systems to **manage** multiple **activities**.
- Allows **point-and-click interaction** for selecting menu items, icons, and objects.
- **Examples** include word processing, spreadsheet, and desktop-publishing programs.

❖ Plotting:

- Creates **2D and 3D** graphs of mathematical, physical, and economic **functions**.
- **Commonly** used **plots** include histograms, bar charts, pie charts, and task-scheduling charts.
- Helps in **presenting trends and patterns** of complex data concisely.

❖ Office Automation and Electronic Publishing:

- Facilitates **in-house** production of printed (hardcopy) and electronic (softcopy) **documents**.
- Documents can contain text, tables, graphs, and drawn or scanned-in graphics.

❖ Computer-Aided Drafting and Design (CADD):

- Designs components and systems of various mechanical, electrical, and electronic devices.
- Used for **testing** structural, electrical, and thermal **properties** of systems.

► APPLICATIONS OF CG :

◆ Scientific and Business Visualization:

- **Scientific visualization:** generates graphics for scientific, engineering, and medical data.
- **Business visualization:** generates graphics for non-scientific data such as economic data.
- Makes it easier to **understand trends and patterns** in large data sets.

◆ Simulation and Modeling:

- Simulates **real-life conditions** for training and optimization purposes.
- **Examples** include astronaut weightlessness simulators, flight simulators, and military training simulators.

◆ Entertainment:

- Used in the **production** of animated movies, scientific movies, and video games.
- **Examples** include Disney movies like "The Lion King" and video games like FIFA and Doom.

► APPLICATIONS OF CG :

❖ Art and Commerce:

- **Produces pictures** for commercial, scientific, or educational presentations.
- Used in advertising, slide production for presentations, and attracting attention in public spaces.

❖ Cartography:

- Produces **maps and charts** from measurement data.
- **Examples** include geographic maps, oceanographic charts, weather maps, and population-density maps.
- Software like **Surfer** is used extensively for cartographic purposes.

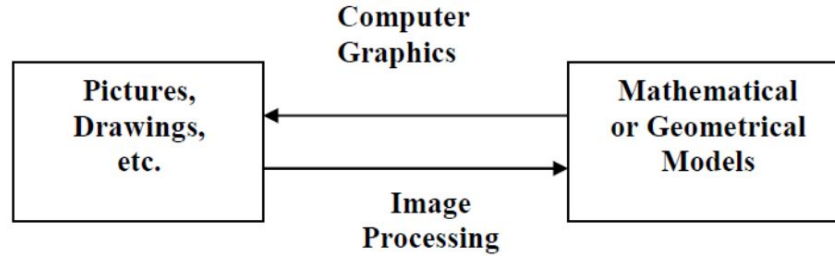
► Advantages of Computer Graphics:

- ❖ **Improved Communication:** Easier to **communicate ideas** through visual representations and interactive learning materials.
- ❖ **Efficient Design and Prototyping:** Facilitates virtual prototyping in engineering and manufacturing, saving time and costs.
- ❖ **Enhanced Visualization:** Realistic images and simulations aid in **better understanding** and visualization of complex data.
- ❖ **Entertainment:** Essential for animated films, special effects, and video games, enhancing **virtual reality experiences**.
- ❖ **Interactive Interfaces:** **Creates user-friendly** and intuitive **interfaces** with real-time feedback.
- ❖ **Precision and Accuracy:** Allows precise control over dimensions, crucial in **medical imaging and research**.

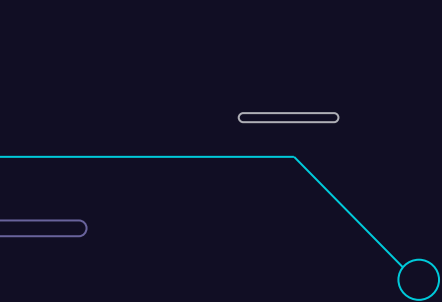
► Disadvantages of Computer Graphics:

- ❖ **High Cost:** Requires expensive hardware, software, and training for skilled personnel.
- ❖ **Resource Intensive:** Demands significant computational power and substantial storage capacity.
- ❖ **Complexity:** Steep learning curve for mastering tools and software with technical challenges.
- ❖ **Time-Consuming:** High-quality rendering and detailed graphics creation can be time-consuming.
- ❖ **Dependence on Technology:** Vulnerable to software bugs, hardware failures, and frequent need for updates.
- ❖ **Ethical and Social Concerns:** Potential for misuse in creating misleading visuals and risk of job displacement.
- ❖ **Health Concerns:** Prolonged screen time can lead to eye strain and poor ergonomic setups can cause repetitive strain injuries.

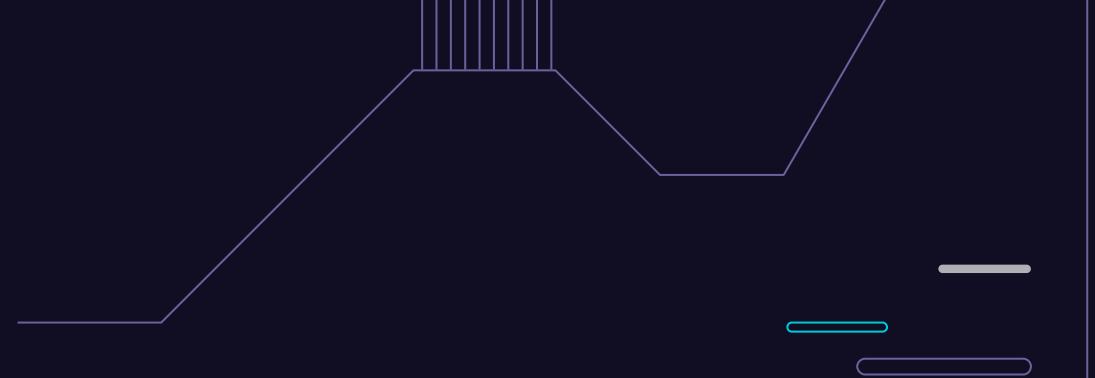
► COMPUTER GRAPHICS vs IMAGE PROCESSING



Computer graphics	Image processing
Computer graphics involves in generating images from mathematical or geometrical models.	Image processing involves in analyzing the images to generate mathematical or Geometrical models.
It includes the creation, storage and manipulation of images or objects.	It is the part of computer graphics that handles image manipulation or interpretation.
E.g. Drawing a picture	E.g. Making blur image visible.



▶ Graphics Hardware

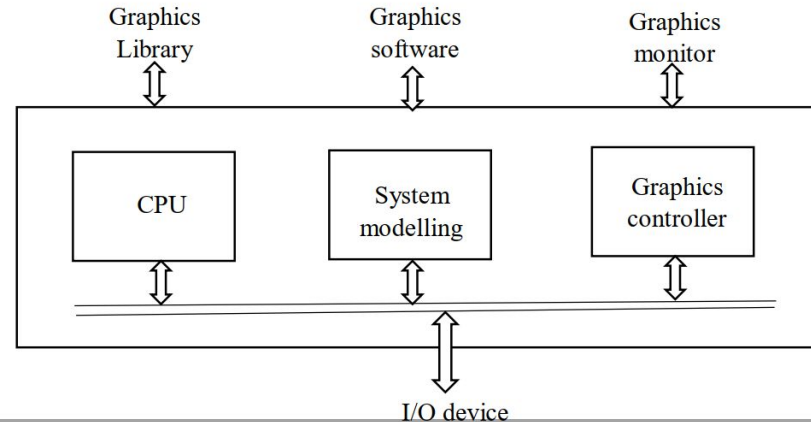


1.2

► Graphics Hardware System:

What is Graphics Hardware System?

- A Graphics Hardware System, also known as a **Graphics Processing Unit (GPU)** or graphics card, is a specialized electronic circuitry component designed to accelerate the rendering of images, videos, and graphical content on a computer display.
- It works in **conjunction** with the computer's central processing unit (CPU) to process and generate visual output.

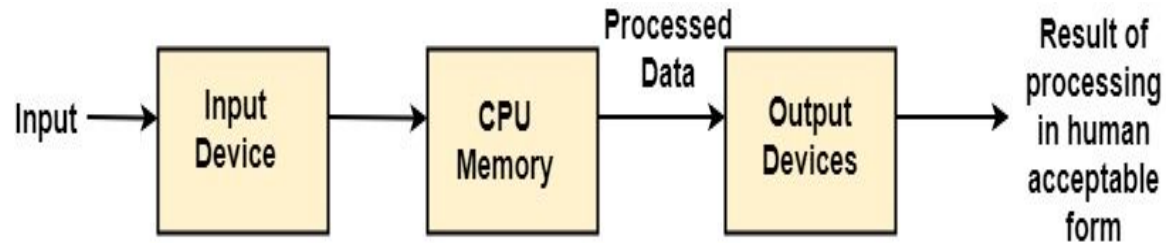


► Hardware ► Input Devices:

- ❖ The Input Devices are the **hardware** that is used to feed input data or information to the computer. The data can be in the form of text, graphics, sound, and text.
- ❖ Data input devices like keyboards are used to provide **additional data** to the computers whereas pointing and selection devices like mouse, light pens, touch panels are used to provide **visual and indication input** to the application.

These Devices include:

1. Tablet
2. Touch panel
3. Light Pen
4. Keyboard
5. Mouse etc



► Hardware ► Input Devices:

1. Tablet

Definition: A digitizer device that scans an object and inputs discrete coordinate positions.

Types:

- **Electrical Tablet:** Uses a grid of wires and electromagnetic signals to determine stylus position.
- **Sonic Tablet:** Utilizes sound waves and microphones to detect the stylus location.
- **Resistive Tablet:** Involves a glass surface coated with a conductive material, with the stylus emitting high-frequency signals.

2. Touch Panel

Function: Allows users to point at the screen directly to move the cursor or select icons.

Types:

- **Optical Touch Panel:** Uses infrared LEDs and photodetectors to form a grid, detecting touch by the interruption of light beams.
- **Sonic Touch Panel:** Employs high-frequency sound waves, calculating touch position based on reflected waves.
- **Electrical Touch Panel:** Consists of two transparent plates that detect touch by measuring contact between conducting and resistive materials.

► Hardware ► Input Devices:

3. Light Pen

Definition: A pencil-shaped device used to determine screen coordinates by detecting light bursts.

- **Drawbacks:** Obscures screen, causes arm fatigue, cannot detect black points, and gives false readings in certain lighting conditions.

4. Keyboard

Function: Generates unique codes (e.g., ASCII) for pressed keys; used for cursor movement, menu selection, and text input.

- **Design:** Often includes alphanumeric keys, function keys, cursor-control keys, and a numeric pad.

5. Mouse

Function: A hand-held device for cursor positioning on the screen.

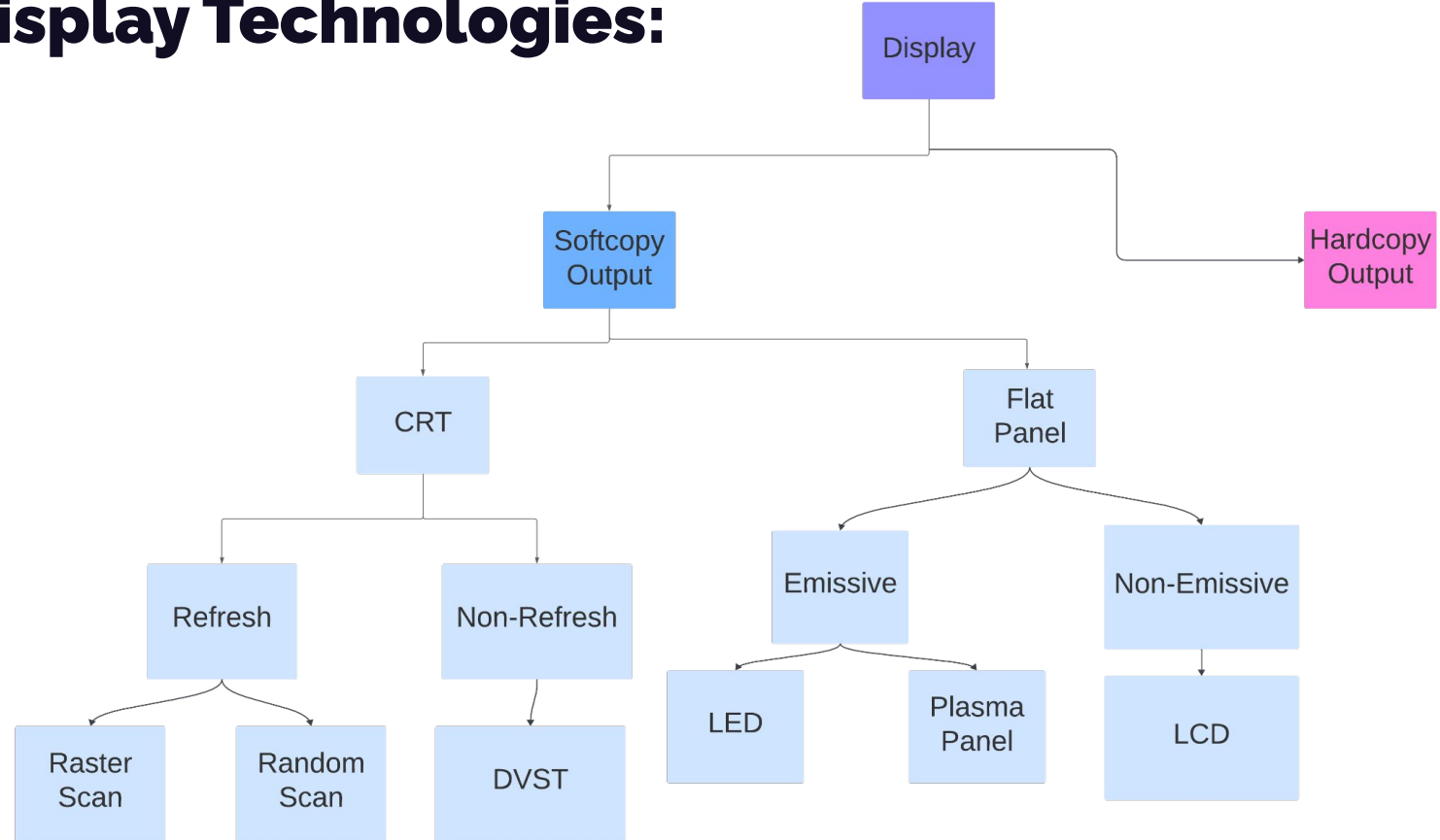
Types:

- **Mechanical Mouse:** Uses a ball and orthogonal wheels to detect movement via light interrupts.
- **Optical Mouse:** Uses a camera and red LED to detect movement based on reflected light patterns.

► Hardware > (**Soft Copy Output , Display Technologies**):

- ❖ Typically, the **primary output device** in a graphics system is a **video monitor**.
- ❖ The operation of most video monitors is based on the **standard cathode-ray tube** (CRT) design, but several other technologies exist and solid-state monitors are predominant.
- ❖ The **display devices** used in graphics system is **video monitor**. The most common video monitor is based on CRT technology.

► Display Technologies:



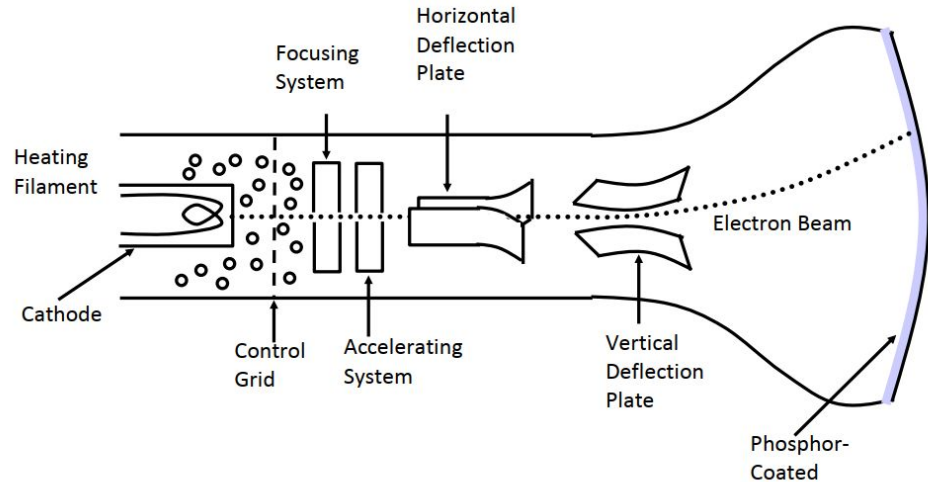
► Cathode Ray Tube (CRT)

❖ Video Display Devices:

Primary output device in a graphics system is **video monitor** whose operation is based mostly on standard cathode-ray tube (CRT) design.

❖ Cathode Ray Tube(CRT):

- A CRT is an evacuated glass tube, with a heating element on one end and phosphor-coated screen on the other end.



► Cathode Ray Tube (Contd...)

Working Mechanism of CRT :

- When a current flows through this **heating element (filament)** the conductivity of metal is reduced due to high temperature. These cause electrons to pile up on the filament.
- These electrons are **attracted** to a **strong positive charge** from the outer surface of the focusing anode cylinder.
- Due to the **weaker negative charge** inside the cylinder, the electrons head towards the anode **forced into a beam** and accelerated towards phosphor-coated screen by the high voltage in inner cylinder walls.
- The forwarding fast electron beam is called **Cathode Ray**.

► Cathode Ray Tube (Contd...)

Working Mechanism of CRT :

- There are two sets of **weakly charged deflection plates**. The first set displaces the beam **up and down** and the second displaces the beam **left and right**.
- The electrons are sent flying out of the neck of bottle (tube) until the **smash into** the phosphor coating on the other end.
- When electrons strike on phosphor coating, the phosphor then **emits a small spot of light** at each position contacted by electron beam. The glowing positions are used to represent the picture in the screen.

Question - How is the brightness of display controlled?

⇒ (controlled by varying the voltage on the control grid)

► Components of CRT:

- ❖ **Electron Gun:** Electron gun consisting of a series of elements, primarily a **heating filament** (heater) and a **cathode**. The electron gun creates a source of electrons which are focused into a narrow beam directed at the face of the CRT.
- ❖ **Control Electrode:** It is used to turn the electron beam **on and off**.
- ❖ **Focusing system:** It is used to **create a clear picture by focusing the electrons** into a narrow beam.
- ❖ **Deflection Plates:** It is used to **control the direction** of the electron beam. It creates an **electric or magnetic field** which will bend the electron beam as it passes through the area.
- ❖ **Phosphor-coated screen:** The inside front surface of every CRT is coated with phosphors. Phosphors **glow when a high-energy electron beam hits them**.

► Important Supporting Terms of CRT :

Fluorescence / Phosphorescence:

- ❖ When the beam of electron emitted by electron gun **strikes phosphor coated screen** on the CRT, the phosphor emits a small spot of light at **each position contacted by the electron beam**, such phenomenon is known as fluorescence / phosphorescence.
- ❖ It last just a fraction of millisecond (1 millisecond = 1/1000 second or 10^{-3} second).

Persistence:

How long a phosphor continues to emit light after the electron beam is removed?

- ❖ Persistence of phosphor is defined as **time it takes for emitted light to decay to 1/10 (10%) of its original intensity**. Range of persistence of different phosphors can be many seconds.
- ❖ The phosphor used for **graphics display device** usually have persistence of **10 to 60 microsecond (10^{-6})**.

► Important Supporting Terms of CRT :

Resolution :

- ❖ Resolution is defined as the **maximum number of points that can be displayed horizontally and vertically without overlap** on display device. More precise definition of resolution is no of dots per inch (**dpi/pixel per inch**) that can be plotted horizontally and vertically.

Horizontal Scan Rate:

- ❖ The horizontal scan rate is the **number of scan lines accessed or visited per second**. The rate is approximately the product of the refresh rate and the number of scan lines.

Aspect Ratio:

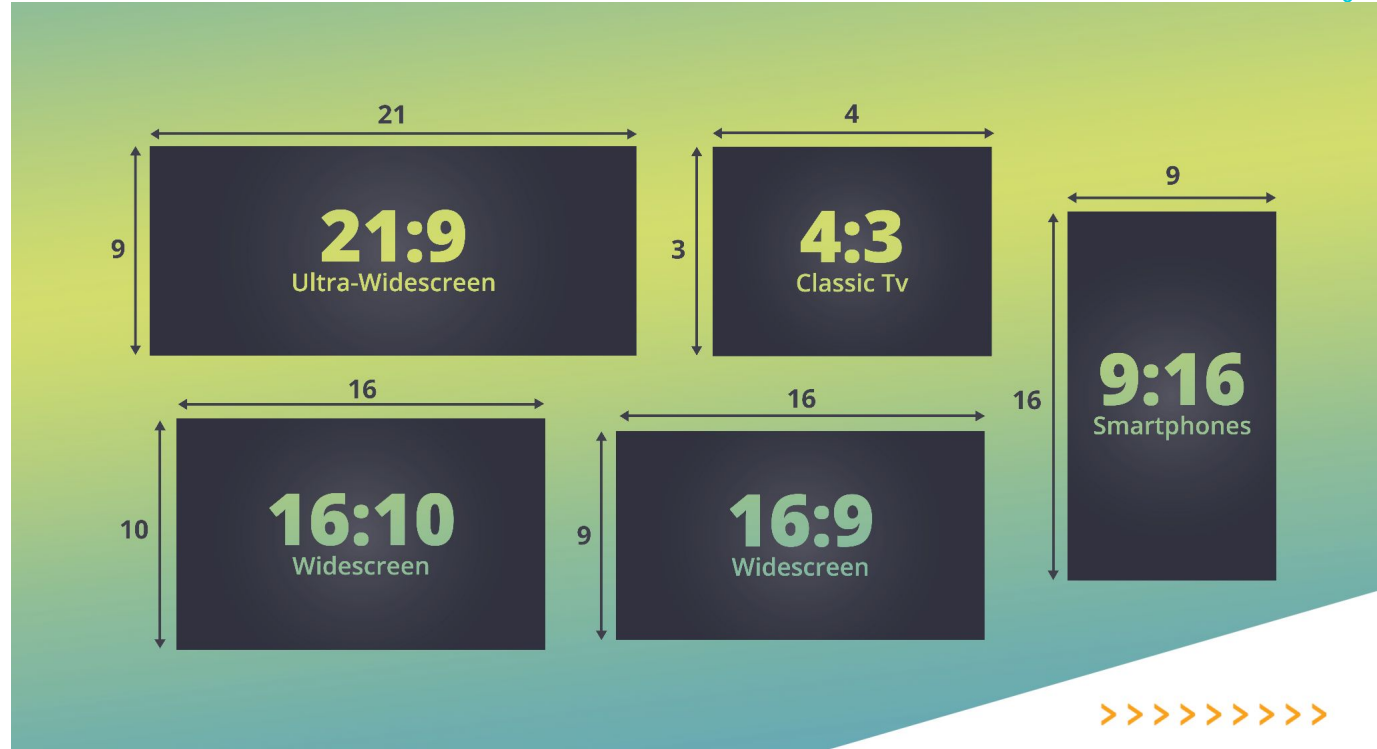
- ❖ It gives the **ratio of vertical point to horizontal point** necessary to produce **equal length lines** in both directions on the screen.
- ❖ An aspect ratio of $\frac{3}{4}$ means that a vertical line plotted with 3 points has the same length as a horizontal line plotted with 4 points.

$$\blacksquare \text{ Aspect Ratio} = \text{Width of image} / \text{Height of image}$$

► Important Supporting Terms of CRT :

Aspect Ratio:

Aspect Ratio =
$$\frac{\text{Width of image}}{\text{Height of image}}$$



► Important Supporting Terms of CRT :

Refresh Rate:

- ❖ Light emitted by phosphor fades very rapidly, so to keep the drawn picture glowing constantly; it is required to **redraw the picture repeatedly and quickly** directing the electron beam back over the same point. The **no of times/sec** the image is **redrawn to give a feeling of non-flickering pictures** is called refresh-rate.

Frame Buffer:

- ❖ Frame Buffer is the **large contiguous memory** into which the intensity values for all the pixel are placed. It contains all the internal representation of the image.
- ❖ It is the video memory that is used to **hold or map the image displayed** on the screen. It generally depends upon the **resolution of the screen image** and **the color depth**.
- ❖ **Formula** to calculate frame buffer is:
 - **Memory MB = (x-resolution * y-resolution * bit per pixel) / 8 * 1024 * 1024**

► Numerical Section:

Q. How much kilobytes does a frame buffer need in $600 * 400$ pixels?

Solution:

Resolution: $600 * 400$

Let n bits are required to store 1 pixel (Assume, because not provided)

Size of Buffer = ?

Now,

Size of Buffer = resolution * bits per pixel

$$= 600 * 400 * n \text{ bits}$$

$$= 240000n \text{ bits}$$

$$= 240000n / 8 * 1024$$

$$= 29.30n \text{ Kilobytes}$$

► Numerical Section:

Q. A system with 24 bits per pixel and resolution of 1024 *1024. Calculate the size of frame buffer.

Solution:

Resolution: 1024*1024

Bits per pixel = 24

Size of Buffer =?

Now,

Size of Buffer = resolution * bits per pixel

$$= 1024 * 1024 * 24 \text{ bits}$$

$$= 25,165,824 \text{ bits}$$

$$= 1024 * 1024 * 24 / 8 \text{ bytes}$$

$$= 1024 * 1024 * 24 / 8 * 1024 \text{ Kilobytes}$$

$$= 1024 * 1024 * 24 / 8 * 1024 * 1024 \text{ Megabytes} = 3\text{MB}$$

► Numerical Section:

Q. Consider a RGB raster system is to be designed using 8 inch by 10 inch screen with a resolution of 100 pixels per inch in each direction. If we want to store 8 bits per pixel in the frame buffer, how much storage do we need for the frame buffer?

Solution:

Size of screen = 8 inch $\times$ 10 inch

Pixel per inch (Resolution) = 100

Total no. of pixel = $(8 \times 100) \times (10 \times 100) = 800000$ pixels

Per pixel storage = 8 bit

Total storage required in frame buffer = 800000×8 bits = 6400000 bits
= $6400000 / 8$ bytes = 800000 byte

► Numerical Section:

Q. Find out the aspect ratio of raster system using 8*10 inches screen and 100 pixel per inch.

Solution:

We know that,

$$\text{Aspect ratio} = \frac{\text{width}}{\text{height}} = \frac{8 \times 100}{10 \times 100} = \frac{4}{5}$$

So, aspect ratio = 4:5

► Numerical Section:

Q. An image has a height of 3 inches and an aspect ratio of 2.5. Find the width of image.

Solution:

Width of image = ?

Height of image = 3 inches

Aspect Ratio = 2.5

Now,

Aspect Ratio = Width of image / Height of image

Width of image = AR * Height of Image

$$= 2.5 * 3$$

$$= 7.5 \text{ inches}$$

► Numerical Section:

Q. A computer has display that is 21.5 inches with 1920*1080 resolution. Find the pixel density.

Solution:

Display Size = 21.5 inches

Width of image = 1920 & Height of image = 1080

Pixel Density = ?

Now,

Pixel Density = Diagonal Resolution / Display Size

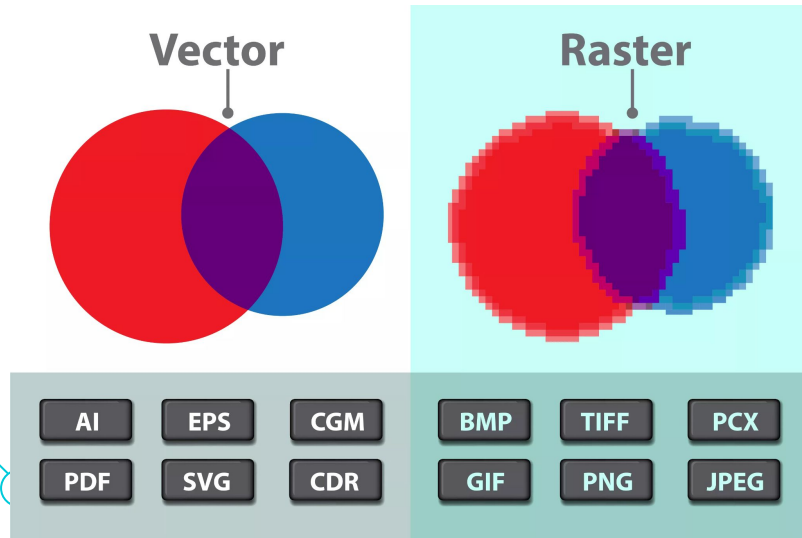
Diagonal Resolution = $\sqrt{1920^2 + 1080^2} = 2203$

Pixel Density = $2203 / 21.5$
= 103 ppi

► Raster vs Vector Graphics:

Raster Graphics	Vector Graphics
Raster graphics are composed of pixels.	Vector graphics are composed of paths.
Raster image pixels do not keep on their appearance as size increases- when you blow a photograph up, it becomes blurry for this reason.	Vector images keep on appearance regardless of size, since the mathematical formulas dictate how the image is rendered.

► Raster vs Vector Graphics:



VECTOR

VS

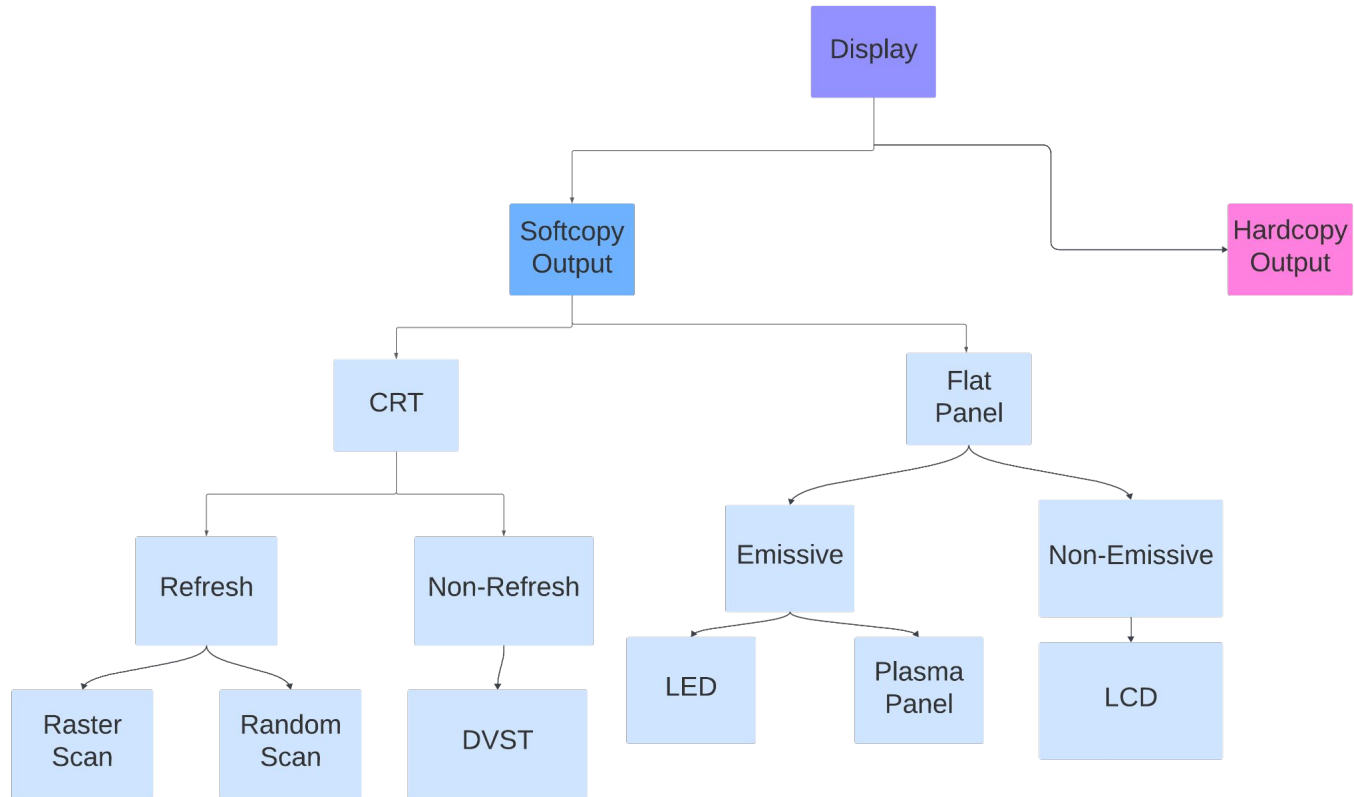


RASTER



▶ **Raster Scan Display System, Video Controller, Random Scan Display System**

► Display Technologies:



► Raster Scan Display

The most common type of graphics monitor employing a CRT is the raster-scan display, based on television technology.

Working Mechanism:

- In raster-scan the electron beam is swept across the screen, **one row** at a time from **top to bottom**. No of scan line per second is called **horizontal scan rate**.
- As electron beam moves across each row, the beam **intensity is turned on and off** to create a pattern of illuminated spots
- **Picture definition** is stored in a memory called frame buffer or refresh buffer. Frame buffer holds all the **intensity value** for screen points. Stored intensity values are then **retrieved** from the frame buffer and “painted” on the screen **one row** (scan line) at a time.
- Returning of electron beam from **right end to left** end after **refreshing each scan line** is called horizontal retrace.
- At the **end** of each frame, the electron beam returns to the **top left corner** to begin next frame called vertical retrace.
- Availability of frame buffer makes raster-scan display well suited for the realistic display.
- Example: Monitors, Home television, printers.

► Raster Scan Display (Contd...)

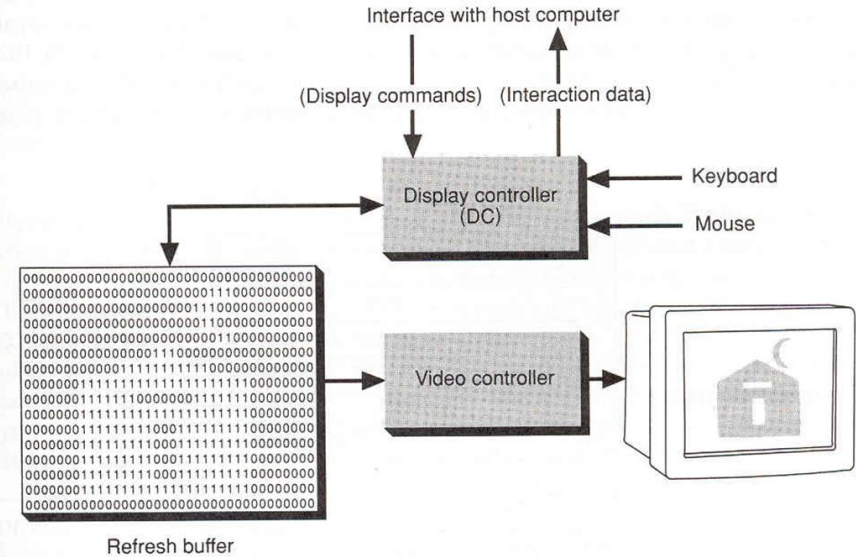
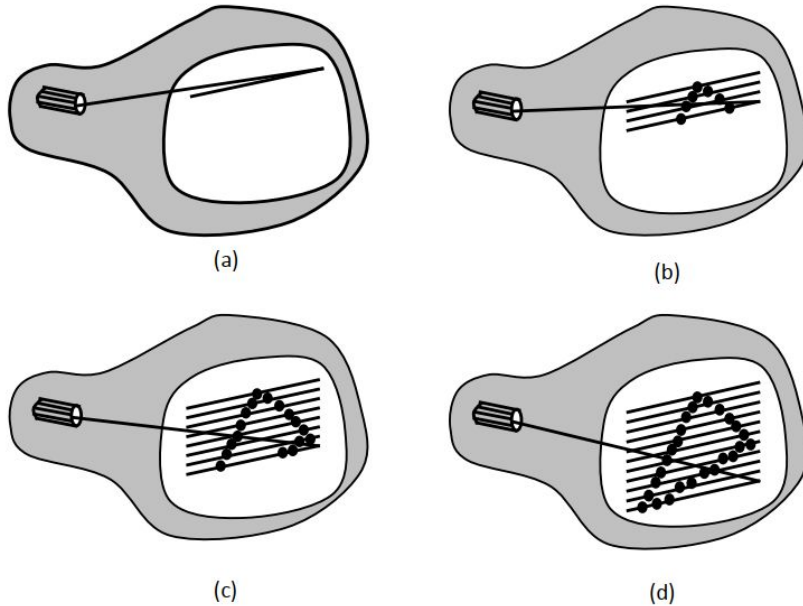


Figure: A raster-scan system displays an object as a set of points across each screen scan line

► Raster Scan Display (Contd...)

Pixel position depends on capability of raster system.

- For **B/W system** each point on screen is either on or off, so only one bit per pixel is needed to control the pixel intensity.
- To display color with varying intensity level, additional bits are needed.

Up to 24 to 32 bit per pixel are included in high quality systems, which require more space of storage for the frame buffer, depending upon the resolution of the system.

Example: A system with 24 bit pixel and screen resolution 1024*1024 require 3 megabyte of storage in frame buffer.

$$1024*1024 \text{ pixels} = 1024*1024*24 \text{ bits} = 3 \text{ MB (using 24-bit per pixel)}$$

The frame butter in B/W system stores a pixel with one bit per pixel so it is termed as **bitmap**. The frame buffer in multi bit per pixel storage is called **pixmap**.

Refreshing on Raster-Scan display is carried out at the **rate of 60** or higher frames per second.

► Raster Scan Display (Contd...)

Refreshing on Raster-Scan display is carried out at the rate of 60 or higher frames per second.

Refresh Procedure:

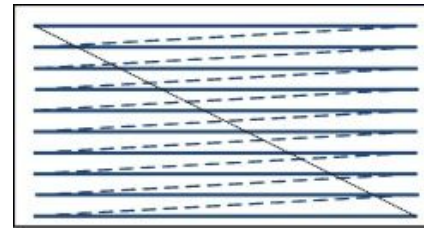
- In **interlaced refresh procedure**, each frame is displayed in two passes. First pass for odd scan lines and another for even ones.
- In **non-interlaced refresh procedure**, electron beam sweeps over entire scan lines in an frame from top to bottom in one pass.



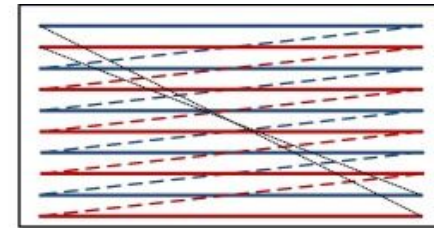
Interlaced



Non-interlaced



Progressive scan



Interlaced scan

— Field 1 (odd)
- - Field 2 (even)

► Raster Scan Display (Contd...)

Refresh Procedure:

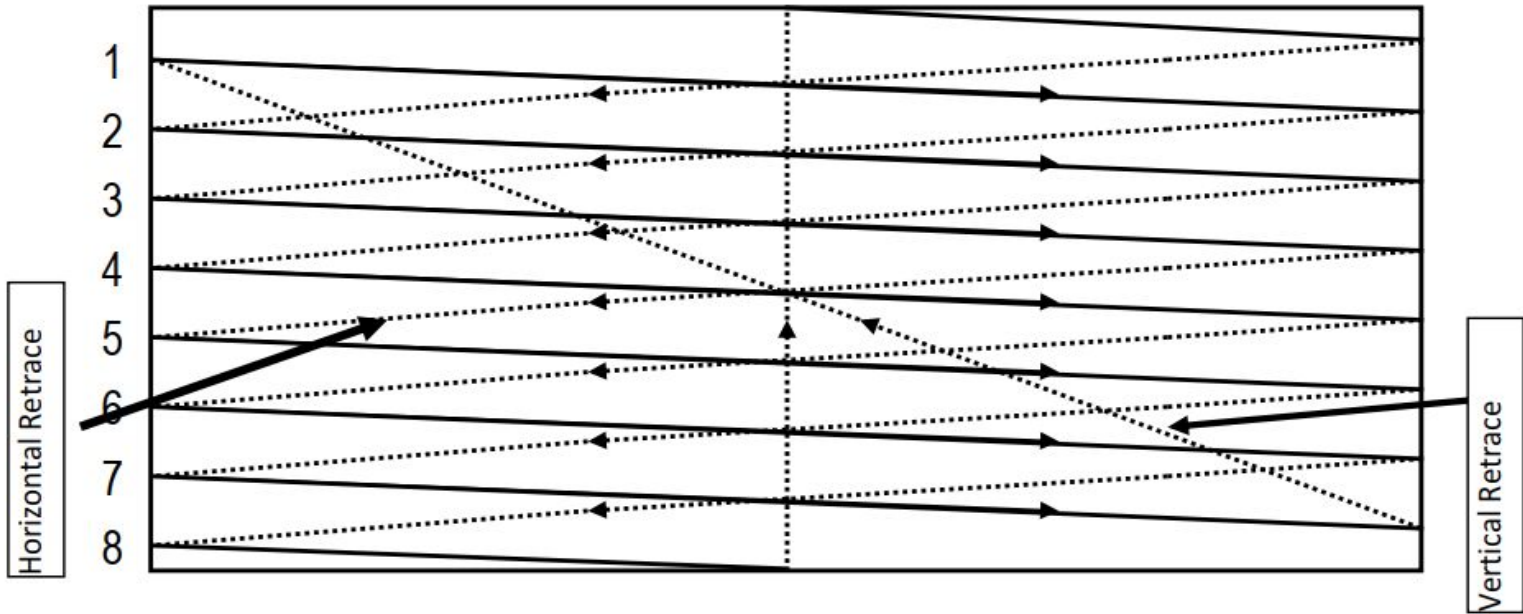


Figure: Horizontal retrace and Vertical retrace

► Raster Scan Display (Contd...)

Advantages:

- ❖ It has an ability to fill the areas with solid colors or patterns
- ❖ The time required for refreshing is **independent** of the complexity of the image
- ❖ Low cost

Disadvantages:

- ❖ Its resolution is **poor**.
- ❖ For Real-Time dynamics not only the end points are required to move but all the pixels in between the moved end points have to be scan converted with appropriate algorithms which might **slow down the dynamic process**.

► Raster Scan Display Architecture:

- There is a special purpose processor called **video controller** or display controller, is used to **control the operation** of the display device.
- When a particular **command** is called by the application program, the graphics **subroutine** package sets the appropriate pixels in the **frame buffer**. The video controller then cycles through the frame buffer, one scan line at a time. It will bring a value of each pixel contained in the frame buffer and uses it to **control the intensity** of the CRT electron beam.
- The display processor is a separate processor that **performs graphics function** such as scan conversion and raster operation.
- System memory holds data and those program that execute on the CPU.
- The display processor memory holds **data plus program** that perform scan conversion and raster operation.
- The frame buffer stores displayable image created by scan conversion & raster operation.

► Raster Scan Display Architecture:

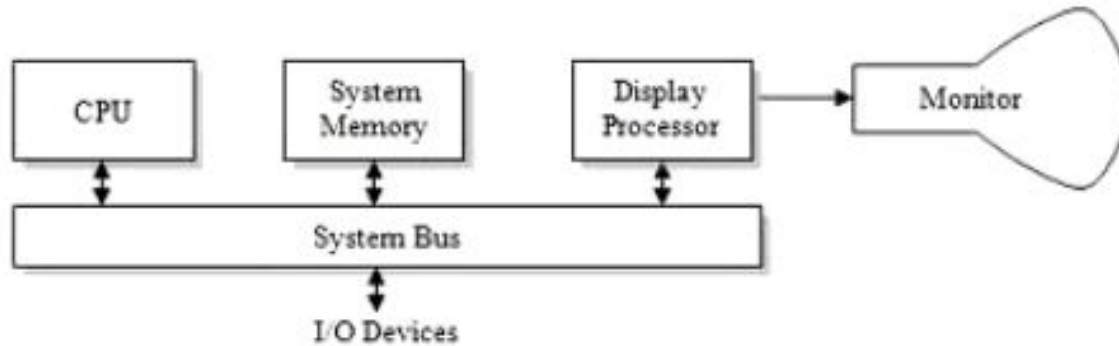
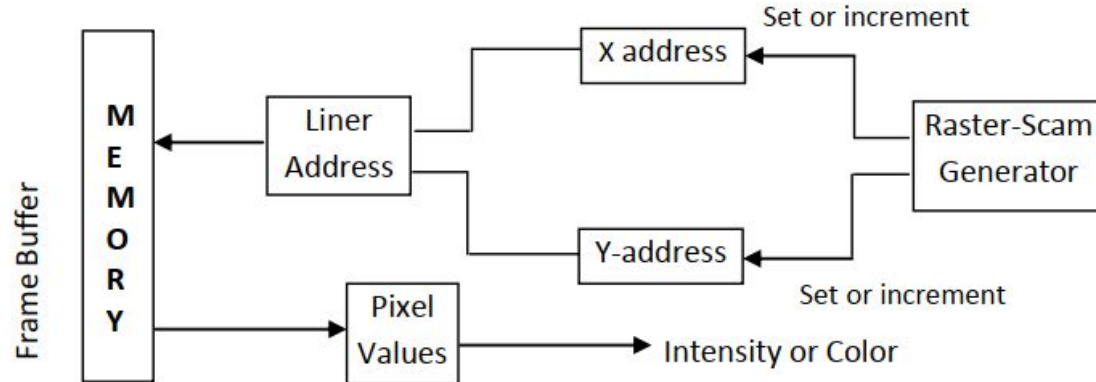


Fig: Architecture of simple raster-graphics system

► Video Controller Architecture:

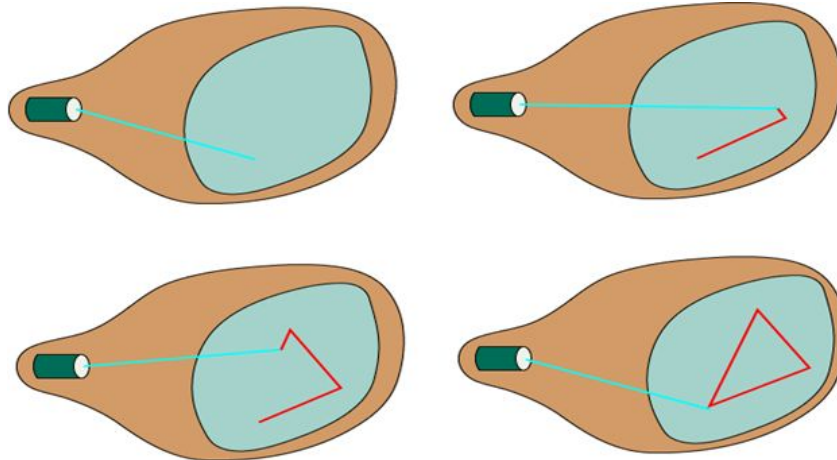
- It is a **special-purpose processor** used to control the operation of the display device.
- **Two Registers** (x and y) are used to **store screen pixel coordinates**. Initially, $x = 0$ and $y = 0$
- As first scan line is generated, the x register is incremented up to x_{max} . Each pixel value is fetched and used to control the intensity of CRT beam. After first scan line, x register address is reset to 0 and y register address is incremented by 1. The process is continued until the last scan line ($y = y_{max}$) is generated.



► Random Scan (Vector) Display

Working Mechanism:

- In random scan system, the electron beam is directed **only to the part of screen** where the picture is to be drawn. It draws a picture one line at a time, so it is also called vector display.
- **Picture definition** is stored as a **set of line drawing commands** in an area of memory called refresh display file.
- To display a picture, the system cycles through the set of commands in the display file. After all commands are processed, the system cycle backs to the first line command in the list.
- The refresh rate of vector display depends upon the no of lines to be displayed for any image.



► Random Scan (Vector) Display Architecture:

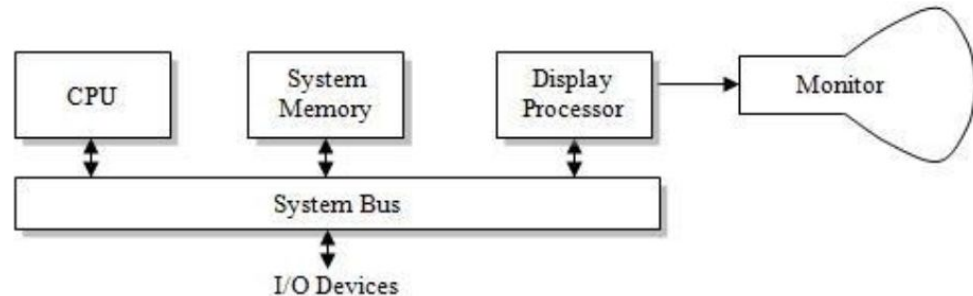
- Vector display system consists of several units along with peripheral devices. The display processor is also called as graphics controller.
- Graphics package creates a display list and stores in systems memory (consists of points and line drawing commands) called display list or display file.
- Vector display technology is used in monochromatic or beam penetration color CRT.
- Graphics are drawn on a vector display system by directing the electron beam along component line

Advantages:

- Can produce output with high resolutions.
- Better for animation than raster system since only endpoint information is needed.

Disadvantages:

- Cannot fill area with pattern and manipulate bits.
- Refreshing image depends upon its complexity.



► Raster Scan System vs Random Scan System:

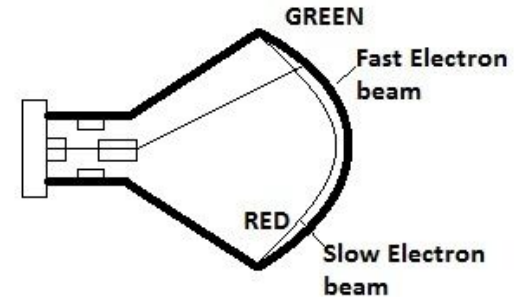
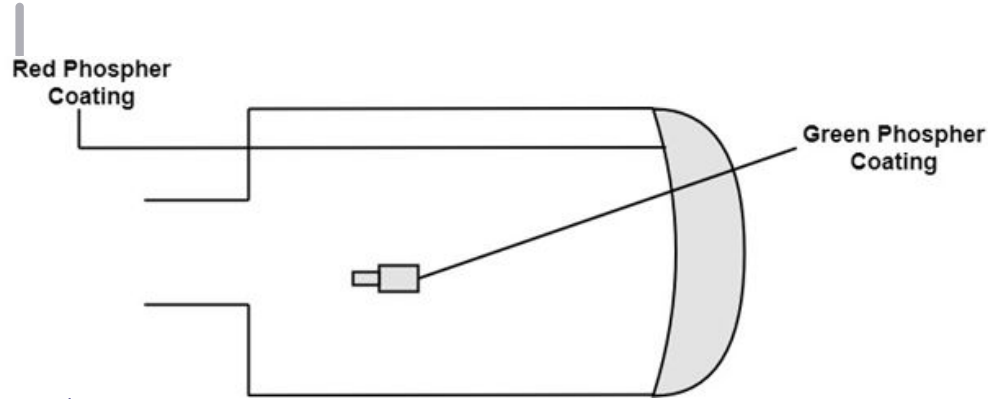
Raster scan system	Random scan system
The electron beam is swept across the screen, one row at a time, from top to bottom.	The electron beam is directed only to the parts of screen where a picture is to be drawn.
Its resolution is poor.	Its resolution is good.
Picture definition is stored as a set of intensity values for all screen point.	Picture definition is stored as a set of line drawing instructions in a display file.
Screen points / pixels are used to draw image.	Mathematical functions are used to draw an image.
Used in system to display realistic images.	Cannot draw realistic shaded scenes.
Cost is low.	Cost is high.
Graphics primitives are specified in terms of their endpoints and must be scan converted into their corresponding pixels in the frame buffer.	Scan conversion is not required.

► Color CRT Display:

A CRT monitor displays color pictures by using a combination of **phosphors** that emit different colored light. By combining the emitted light from the **different phosphors**, a range of colors can be generated.

Two basic techniques for producing color displays with CRT are:

1. Beam-penetration method
2. Shadow-mask method



► Color CRT Display:

Beam Penetration method:

This method is commonly used for **random scan system**.

In beam penetration method, **two layers** of phosphor usually **red and green** are coated on the CRT screen, and displayed color depends on **how far** the electron **beam penetrates** into phosphor layer.

- **Slow electrons** excite only **red layer** so that we can see red color displayed on the screen pixel where the beam strikes.
- **Fast electrons** beam excite **green layer** penetrating the red layer and we can see the green color displayed at the corresponding position.
- At **Intermediate beam** speeds, combinations of **red and green light** are emitted to show two additional colors - **orange and yellow**. **(4 colors only possible)**
- The speed of the electrons and hence the screen color at any point, is controlled by the **beam-acceleration voltage**

Limitations:

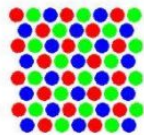
- Quality of picture is **not good** as with other methods.
- It is an **inexpensive** way to produce color in random scan monitors, but only four colors are possible.

Color CRT Display:

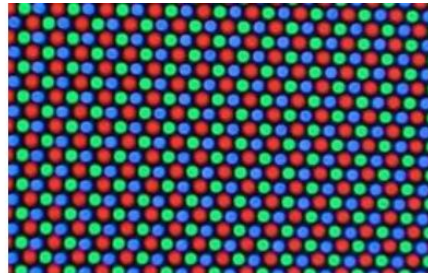
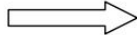
Shadow Mask Method:

- This method is commonly used in **raster scan systems** because they can produce wide range of colors than beam penetration.
- A shadow mask CRT has **three phosphor color dots at each pixel position**. One emits **red** light, another emits **green** light and third emits **blue** light.
- The CRT has **three electron guns**, one for each color dot and a special **metal plate** called **shadow-mask grid** is placed just behind the phosphor coated screen.
- Shadow mask contains a sequence of holes aligned with the phosphor-dot patterns to activate only its corresponding color dot.
- The phosphor on the face of the screen is laid out in a precise geometric pattern. There are two primary variations.
 1. The **stripe pattern** of inline tube (In-line Arrangement)
 2. The **delta pattern** of delta tube (Delta-delta Arrangement)

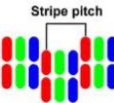
Delta Pattern



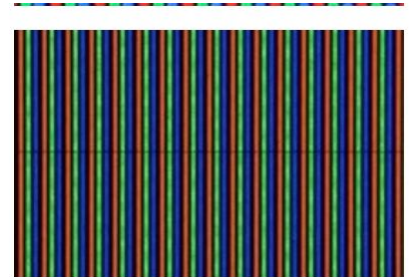
Monitor view



Stripe Pattern

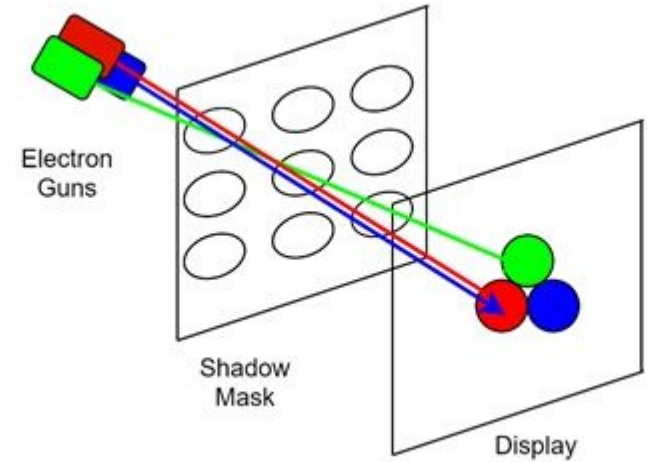
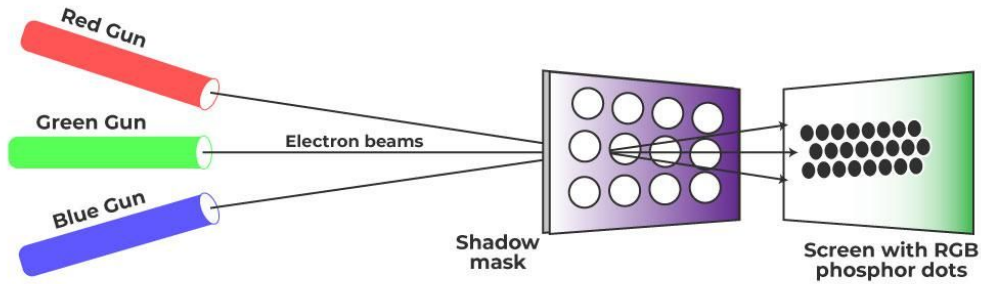


Monitor view



► Color CRT Display:

Shadow Mask Method:



► Direct-View Storage Tubes (DVST):

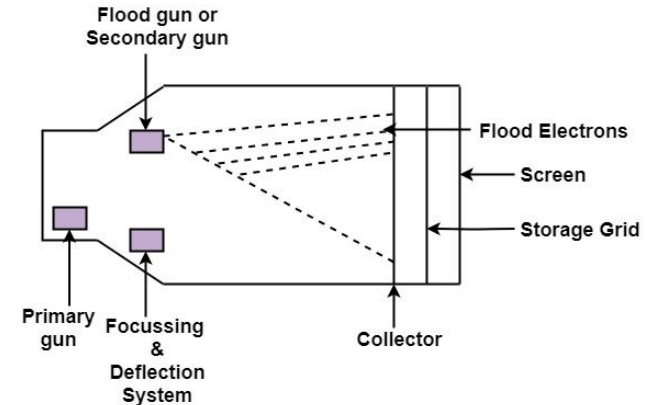
- This is alternative method for method maintaining a **screen image** to store picture information inside the CRT **instead of refreshing** the system.
- DVST stores the picture information as a **charge distribution** just behind the phosphor coated screen.
- **Two electron guns** are used in DVST:
 1. **Primary gun:** Used to store the picture pattern.
 2. **Flood gun:** Used for maintaining the picture display.

Advantages:

- No refreshing is needed.
- Very complex picture can be displayed at very high resolutions without flicker.

Disadvantages:

- To update any part of image must redraw all parts of image.
- Ordinarily, **do not** display color.
- To erasing and redrawing process can take several second for a picture.



Direct View Storage Tube

► Flat Panel Display:

- Flat-Panel Display refers to a class of video devices that have **reduced volume, weight** and **power** requirements compared to CRT.
- These emerging display technologies tend to **replace** CRT monitors. Current uses of flat-panel displays include TV monitors, calculators, pocket video games, laptops, displays in airlines and ads etc

Two categories of flat-panel displays:

1. **Emissive Displays (or emitters):** Emissive displays are device that convert **electrical energy into light**. E.g. Plasma panels, light emitting diode (LED) etc.
2. **Non-emissive Displays (or non-emitters):** Non-emissive displays use **optical effects** to convert **sunlight or light** from other sources into **graphics pattern**. E.g. Liquid Crystal Display (LCD).

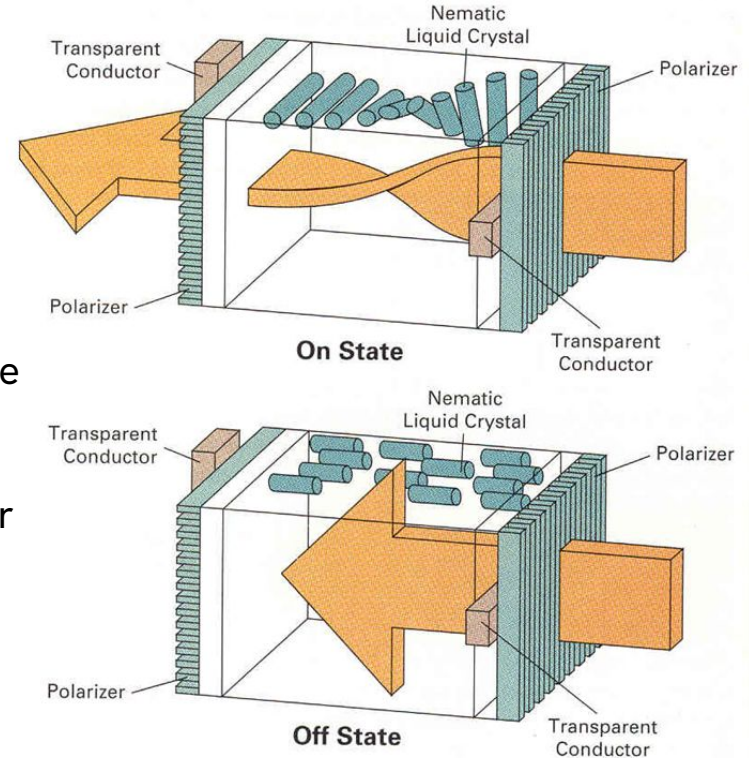
Liquid Crystal Display (LCD): A LCD is a thin, flat display device made up of any **number of pixels arrayed in front of light source or reflector**. It uses very small amounts of electric power, and is suitable for use in battery-powered electronic devices.

For details of flat displays, read page no. 65-67 of book "Computer graphics C version", Hearn & Baker.

► Flat Panel Display:

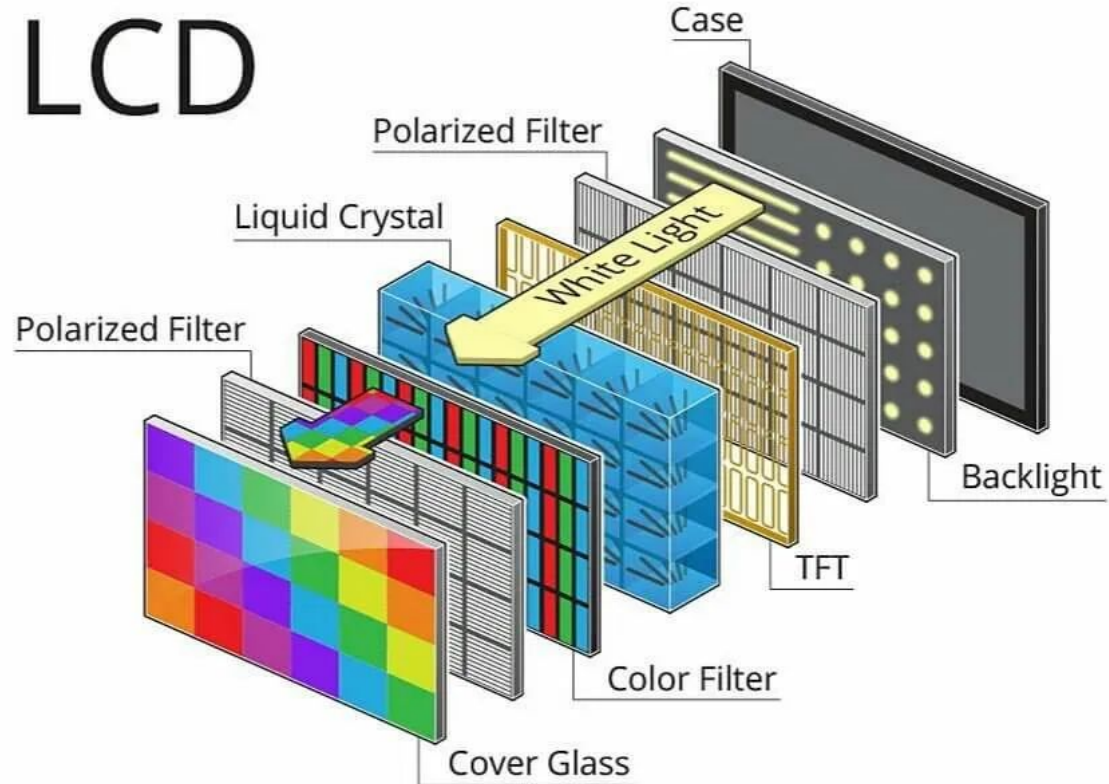
Liquid Crystal Display

- ❖ A transmissive technology
- ❖ Works by letting varying amounts of a fixed-intensity white backlight through an active filter
- ❖ Organic crystals that align themselves together
- ❖ When external force is applied they realign themselves
- ❖ This is used to change polarisation and filter light



► Flat Panel Display:

LCD



Numerical Section

01

► Numerical Section:

Q. Consider raster system with resolution 1280 x 1024.

- a) How much pixel could be accessed per second by the video controller that refreshes the screen at the rate of 60 frames / second?
- b) What is the access time per pixel?

Solution:

a) No. of pixel accessed per second = $1280 * 1024 * 60 = 78643200$ pixels

b) Since 78643200 pixels are accessed in 1 second

$$\text{Access time per pixel} = \frac{1}{78643200} = 12.7 \text{ nanosec.}$$

► Numerical Section:

Q. . Consider a raster scan system having 12 inch by 12 inch with a resolution of 100 pixels per inch in each direction. If display controller of this system refresh the screen at the rate of 50 frames per second, how many pixels could be accessed per second and what is the access time per pixel of the system.

Solution:

Size of screen = 12 inch $\times$ 12 inch

Resolution = 100 pixels per inch

Total no. of pixels in one frame = $(12 \times 100) \times (12 \times 100)$

Refresh rate = 50 frames per second

i.e. 50 frames can be accessed in 1 sec.

Total no. of pixel accessed in 1 sec = $50 \times (12 \times 100) \times (12 \times 100) = 72000000$ pixels

Again,

50 frames can be accessed in 1 sec.

1 frames can be accessed in $1/50$ sec.

$(12 \times 100 \times 12 \times 100)$ frames can be accessed in $1/50$ sec.

Then, 1 pixel can be accessed in $1/(50 \times 12 \times 100 \times 12 \times 100)$ sec.

$$= 10^9 / (50 \times 12 \times 100 \times 12 \times 100) \text{ ns}$$

$$= 13.88 \text{ ns.}$$

► Numerical Section:

Q.What is the time required to display a pixel on the monitor of size 1024×768 with refresh rate of 60 Hz?

Solution:

Refresh rate = 60Hz i.e. 60 frames per second

Total no. of pixel in one frame = $1024 * 768 = 786432$ pixels

60 frames need 1 sec.

1 frame need $1/60$ sec.

786432 pixels need $1/60$ sec.

1 pixel need $1/(60 * 786432)$ sec. = $10^9 / (60 * 786432)$ ns = 21.19 ns

► Numerical Section:

Q. How much time is spent scanning across each row of pixels during screen refresh on a raster system with resolution 1280×1024 and refresh rate of 60 frames per second?

Solution:

Resolution = 1280×1024

i.e. one frame contains 1024 scan line and each scan line consists of 1280 pixels.

Refresh rate = 60 frames per second

i.e. 60 frames take 1 second.

1 frame takes $1/60$ second.

i.e. 1024 scan line take $1/60$ second i.e. 0.0166 sec.

1 scan line take $0.0166/1024 = 0.0000162 \text{ sec}$

► Numerical Section:

Q. If a pixel is accessed from the frame buffer with an average access time of 300ns then will this rate produce an un-flicking effect for the screen size of 640x480.

Solution:

Size of screen = 640×480

Total no. of pixels = $640 \times 480 = 307200$

Average access time of one pixel = 300 ns

Total time required to access entire pixels of image in the screen = $307200 \times 300 = 92160000 \text{ ns}$
 $= 92160000 / 10^9 \text{ sec}$
 $= 0.09216 \text{ sec}$

i.e. one cycle take 0.09216 sec.

Now, no. of cycles per second i.e. refresh rate = ?

$0.09216 \text{ sec} = 1 \text{ cycle}$

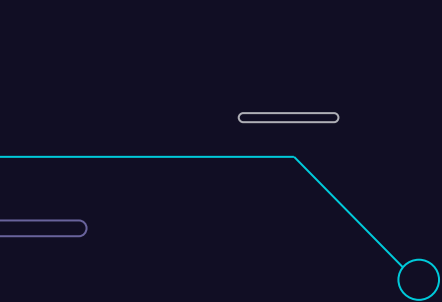
$1 \text{ sec} = 1 / 0.09216$
 $= 10.86$

Refresh rate = 10.86 cycles per second.

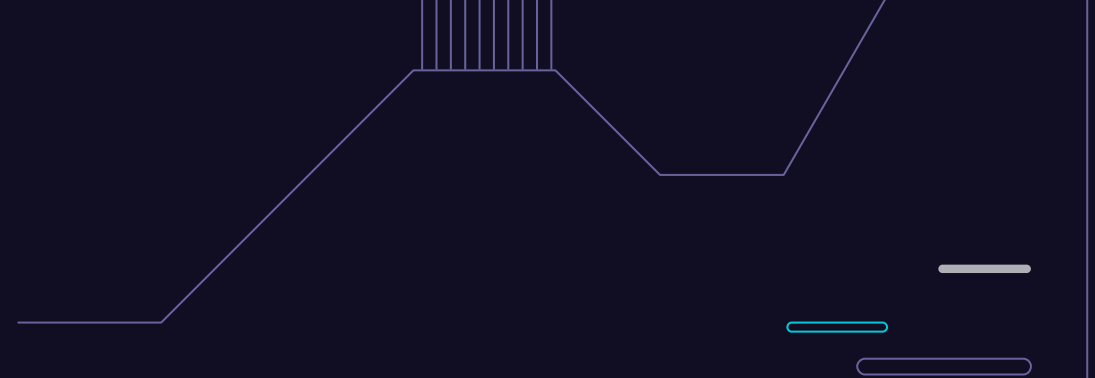
Since the minimum refresh rate for unflicker image is 60 frames per second, hence we can say the monitor produces flickering effect.

► Numerical Section:

Q. . Calculate the total memory required to store a 10 minute video in a SVGA system with 24 bit true color and 25 fps.



▶ Graphics Software



1.3

► Graphics Software:

Graphics software or image editing software is a program or collection of programs that enable a person to manipulate visual images on a computer.

There are two general categories of graphics software:

General programming packages:

- Provides extensive set of graphics functions for high level languages (FORTRAN, C etc).
- Basic functions include those for generating picture components (straight lines, polygons, circles, and other figures), setting color and intensity values, selecting views, and applying transformations.
- Example: OpenGL(Graphics Library)

Special-purpose application packages:

- Designed for non programmers, so that users can generate displays without worrying about how graphics operations work.
- The interface to the graphics routines in such packages allows users to communicate with the programs in their own terms.
- Example: artist's painting programs and various business, medical, and CAD systems.

► Graphics Software Standards:

The primary graphics software standard commonly referred to in the industry is OpenGL (Open Graphics Library). OpenGL is an open-source, cross-platform API (Application Programming Interface) used for rendering 2D and 3D vector graphics. It provides a set of functions for rendering geometric primitives, textures, and images, as well as for performing transformations, lighting, and shading operations.

Importance:

- **Interoperability:** Standards ensure different software can work together without issues.
- **Portability:** Applications can be easily moved across different platforms.
- **Compatibility:** Users can expect consistent performance across software implementations.
- **Innovation:** Standards provide a foundation for building new features and technologies.
- **Accessibility:** Standards make graphics technology more widely available.

► Graphics Software Standards:

Standards:

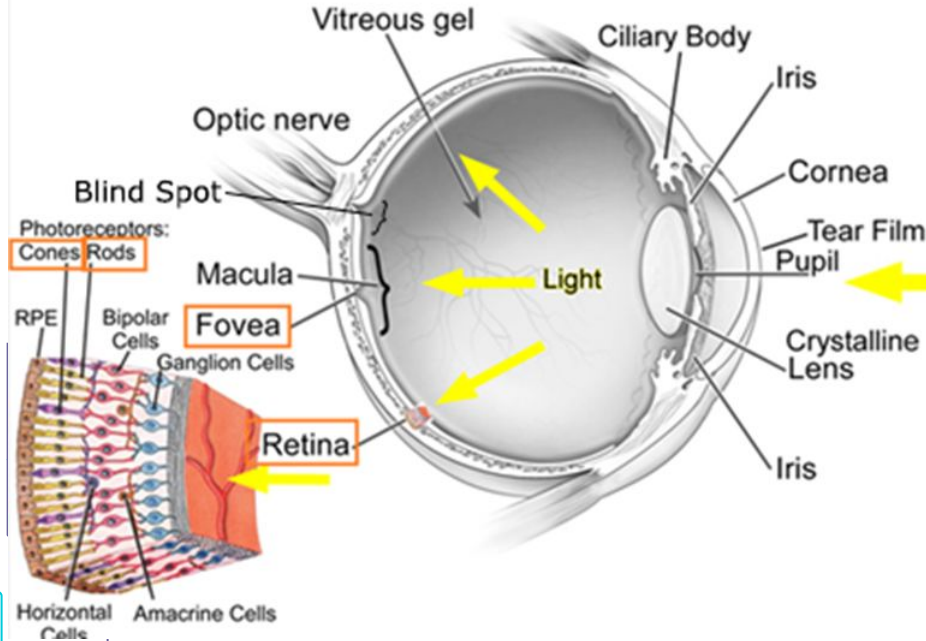
- **GKS (Graphical Kernel System):** First graphics software standard adopted by ISO and ANSI, initially for 2D graphics but later expanded to 3D.
- **PHIGS (Programmer's Hierarchical Interactive Graphics Standard):** Built upon GKS, providing enhanced capabilities for object modeling and surface rendering, later extended by PHIGS+ for advanced 3D shading.
 - **CGI (Computer Graphics Interface):** Standardizes device interfaces for graphics output.
 - **CGM (Computer Graphics Metafile):** Defines standards for archiving and transporting pictures.

► NEED OF MACHINE INDEPENDENT GRAPHICAL LANGUAGE :

The need for a machine-independent graphical language is crucial because:

1. **Portability:** Ensures graphics applications run on diverse hardware without major modifications. For instance, OpenGL allows games to run on both Windows PCs and macOS systems.
2. **Interoperability:** Enables compatibility between different systems, fostering collaboration. SVG (Scalable Vector Graphics) files can be viewed consistently across web browsers regardless of the underlying operating system.
3. **Standardization:** Establishes common conventions, enhancing software development. HTML5 Canvas provides a standardized way to draw graphics in web browsers.
4. **Accessibility:** Makes graphics technology available to a wider audience. Processing, a language based on Java, simplifies graphics programming for artists and designers.
5. **Future-proofing:** Adapts to evolving hardware technologies. Vulkan, a graphics API, supports modern GPUs across various platforms, ensuring graphics applications remain relevant.

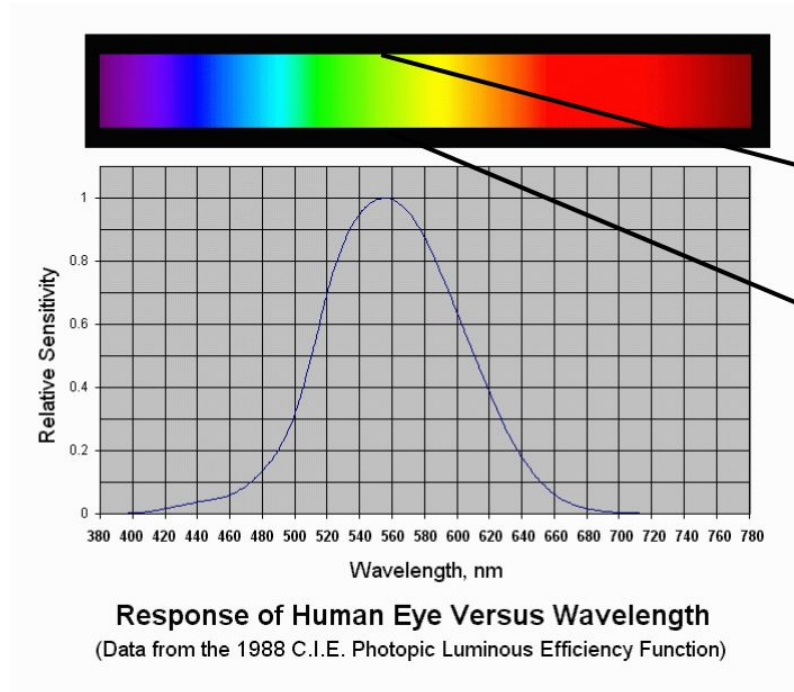
► Human Visual System:



- ❖ Eye "Sees", Brain interprets
- ❖ The human eye has **two types** of detectors
 - Rods and cones
- ❖ **Rods**, basically, only detect **luminance** and are the dominant detector in low light
- ❖ **Cones** detect **color**
- ❖ Three types of **cones**:
 - red, green, and blue (more or less)

► Human Visual System:

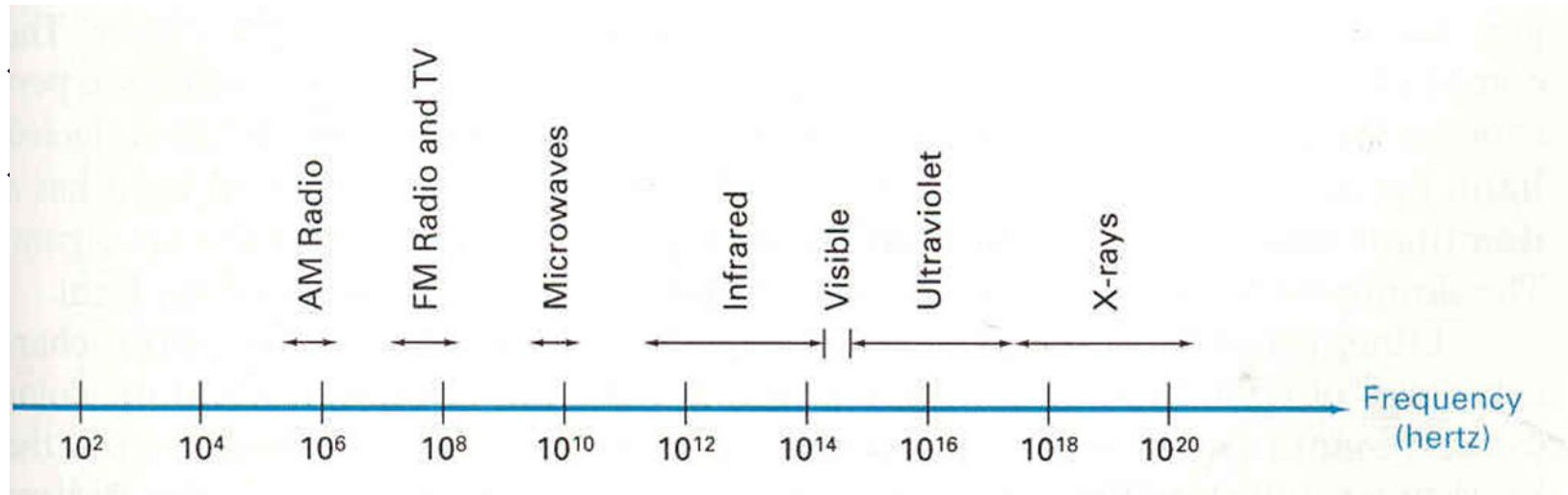
- ❖ To humans (with normal vision), green appears much brighter than other colors.



► Human Visual System:

❖ Visible Light Frequencies Range between

- **Red:** 4.3×10^{14} hertz (700nm)
- **Violet:** 7.5×10^{14} hertz (400nm)



► PYQs:

1. Calculate the total memory required to store a 5 minute video in SVGA system with 24 bit true color and 30 fps. **2080**
2. Calculate the total memory required to store a 8 minute video in a SVGA system with 24 bit true color and 25 fps. **2079**
3. Differentiate between raster and vector graphics method. **2079**
4. What is computer graphics? Explain different application areas of computer graphics. (5)
5. Define horizontal retrace and vertical retrace of electronic beam. Explain architecture of random scan display system. (2+3) **2021**
6. Define refresh rate and persistence of monitor. Explain the architecture of random scan display system with block diagram. **2019** (10)

► Assignment-1:

Due Date: 30th November, 2025

1. Calculate the total memory required to store a 5 minute video in SVGA system with 24 bit true color and 30 fps. **2080**
2. Calculate the total memory required to store a 8 minute video in a SVGA system with 24 bit true color and 25 fps. **2079**
3. Differentiate between raster and vector graphics method. **2079**
4. Differentiate between vector and raster graphics.
5. What is computer graphics? Explain different application areas of computer graphics. (5)
6. Define horizontal retrace and vertical retrace of electronic beam. Explain architecture of random scan display system. (2+3) **2021**
7. Define refresh rate and persistence of monitor. Explain the architecture of raster scan display system with block diagram. **2019** (10)
8. Explain software standards in computer graphics.



▶ **THANKS!**

Do you have any questions?

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